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E-270 LIGHTING

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to the site, installation, commissioning and testing of all lighting, welding outlets and convenience outlets in the buildings and areas in accordance with part 4, Subsection 1.2 Summary of Work.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards and the following specific standards:.

2.1 National

- PUIL : Peraturan Umum Instalasi Listrik - Indonesia
(General Regulation of Electrical Installation – Indonesia).
- SPLN : Standard Perusahaan Umum Listrik Negara.
- Departemen Pekerjaan Umum Indonesia : Standard Penerangan Buatan di Dalam Gedung – gedung

2.2 International Electrotechnical Commission

- IEC 60064 : Tungsten filament lamps for domestic and similar general lighting purposes
- IEC 60079 : Electrical apparatus explosive gas atmospheres
- IEC 60081 : Double-capped fluorescent lamps
- IEC 60083 : Plugs and socket-outlets for domestic and similar general use standardized in member countries of IEC
- IEC 60269 : Low-voltage Fuses.
- IEC 60309 : Plugs, socket-outlets and couplers for industrial purposes
- IEC 60432 : Incandescent lamps
- IEC 60529 : Classification of degrees of protection provided by enclosures

IEC 60598	:	Luminaries
IEC 60662	:	High-pressure sodium vapour lamps
IEC 60755	:	General requirements for residual current operated protective devices
IEC 60921	:	Ballasts for tubular fluorescent lamps
IEC 60923	:	Auxiliaries for lamps
IEC 60947	:	Low-voltage switchgear and controlgear

2.3 National Fire Protection Association

NFPA 70	:	National Electrical Code
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3. OPERATING AND DESIGN CONDITIONS

All equipment shall be suitable for indoor and outdoor service and operation in combined cycle power plant located on a tropical climate along the seacoast.

Equipment installed in hazardous areas shall be in accordance with NFPA 70 as specified in Clause 4.1 herein.

4. DETAILED REQUIREMENTS

4.1 General

The Contractor shall provide the lighting system and air craft obstruction lighting system in the areas indicated herein and as described in Part 4, Subsection 1.2.

In general, the selection of lighting luminaries (fixtures) and/or accessories shall be based on types that are manufactured and/or readily available in Indonesia.

Lighting Source and Illumination levels

Lighting source types and minimum illumination levels for various areas shall be as indicated in Table 4.1-1 and/or in accordance with the recommendations of “Standard Penerangan Buatan di Dalam Gedung Gedung” as issued by the Indonesian Department of Public Works.

Table 4.1-1 Lighting Source Type and Intensity

<u>Area</u>	<u>Source Type</u>	<u>Intensity (Lux)</u>
Turbine Room Main Floor	HPS	320
Turbine Room Basement & Mezzanine	HPS*	220
Circulating Water Pump Area	HPS	220
HRSG Room Ground & Mezzanine	HPS*	220
HRSG Room Main Floor	HPS*	220
HRSG Galleries (Indoor)	HPS*	150
Control Rooms	Fluorescent	750
Auxiliary Equipment Room	Fluorescent	320
Switchgear and Electrical		
Equipment Rooms	Fluorescent	320
Cable Rooms	Fluorescent or HPS	55
Diesel Engine Generator and	Enclosed & Gasketed	
Battery Rooms (NFPA, Division 2)	Fluorescent	220
Oil Rooms (NFPA, Division 2)	Enclosed & Gasketed	220
Machine Shop	HPS	650
Indoor Stairs, Aisles and Corridors	Fluorescent	210
Elevators, etc.		
Misc. Indoor Areas and Structures	HPS: Fluorescent for	
	Low Head Room Areas	210
Misc. Outdoor Galleries and Walk ways	HPS	30
Emergency Lighting	(2.5% of regular lighting)	
Hydrogen Plant (NFPA, Division 2)	Explosion Proof	220
	Fluorescent	
Chlorination Plant	Gas Tight Fluorescent	220
Electronic Equipment Rooms	Fluorescent	750
Offices	Fluorescent	750
Conference Rooms	Fluorescent	750
Locker and Wash Rooms	Fluorescent (except	270
	incandescent in showers)	
Store Rooms (General)	Fluorescent	220
Store Rooms (Bins)	Fluorescent	540
Stairs, Aisles and Corridors	Fluorescent	210
Transformer Area Outdoor	HPS	55
Roadway Lighting	HPS	11
Yard Areas	HPS	11
Building Entrances	HPS	30

VAC room (NFPA, Division 2)	Fluorescent	320
Sewage Treatment Plant	HPS	60
Stack Platforms	HPS	110
Electrical Equipment Rooms (If electrical equipment is mounted inside stack)	Fluorescent	320
Substation Area	HPS	55
Central Control Room	Fluorescent	750

*Fluorescent will be used in low mounting height areas.

- Notes:
- The values shown above are minimum maintained average values. A 15% depreciation allowance should be added.
 - Unless otherwise indicated, lighting shall be high pressure sodium (HPS) luminaries for high bay and outdoor lighting and fluorescent luminaries for low bay and indoor lighting.

Power Supply

The lighting system power supply shall be from 400/230 V, 50 Hz, 3 phase, 4 wire sources.

All luminaries shall meet the applicable requirements of IEC 60598 and shall comply with Class I or Class II requirements for protection against electric shock, as defined in IEC 60598.

All outdoor lighting shall be controlled by photocell controllers with manual overriding switches or heavy duty timers. All indoor lighting shall be controlled by wall mounted switches located near the entrance doors to the lighted areas. Multiway switches shall be used as required. Switch enclosures shall be suitable for the area.

All enclosed areas such as electrical equipment rooms, etc. shall have 20% or a minimum of two (2) luminaries, which shall be "on" at all times. These luminaries shall be wired directly to the lighting branch circuit breakers.

Alternate lighting luminaries for an area and multi-lamp luminaries shall be fed from two (2) alternate lighting cabinets so that the tripping of a lighting cabinet main breaker shall not result in the loss of over 66% of the lighting in an area.

Lighting equipment in hazardous areas, such as the hydrogen plant building (if applicable), battery rooms, diesel engine generator room and oil storage areas, shall meet the requirements of NFPA 70 pertaining to the hazardous substance present in a given area. Hydrogen plant building shall be designated as a class 1 Division 2 location as per NFPA 70. All equipment shall be accessible for maintenance from walkaways or platforms when possible.

4.2 Location /Area Lighting

4.2.1 General

Location/area lighting requirements shall be as specified herein.

Luminaries in sheds and other areas not completely enclosed (i.e. without walls) shall be weather-proof with a degree of protection of IPW55 or better.

4.2.2 Central Control Room

The central control room shall be provided with a louver-all ceiling. The ceiling shall conceal all ventilating equipment and lighting luminaries. The ceiling louver's type, arrangement and installation shall be such that it will not cause any glare on the control board and panels or devices mounted thereon. The lighting shall be fluorescent luminaire strips, as required, to provide the required illumination level indicated in Table 4.1-1. The illumination level shall be adjustable by dimmers.

4.2.3 Road and Yard Lighting

Luminaries for road and yard lighting shall be operated at 400 volts.

It is preferred that the luminaries for yard lighting be mounted on high masts. The high masts shall be provided with a winch arrangement or other method for lowering the luminaries to grade level for revamping and ballast replacement..

4.3 Fluorescent Luminaries

Ballasts for fluorescent luminaries shall be suitable for rapid start lamps in accordance with requirements of IEC 60081.

Ballasts shall be corrected to a minimum power factor of 0.9 lagging and shall be internally protected against short circuits and overheating.

Multiple lamp luminaries shall be provided with individual ballasts for each lamp.

4.4 High Pressure Sodium (HPS) Luminaries

Ballasts for HPS luminaries shall be internally fused and of the constant wattage type. The ballasts shall be corrected to a minimum power factor of 0.9 lagging and shall have a life expectancy of at least 10 years. The ballasts shall meet or exceed the applicable tests specified in IEC 60921.

HPS luminaire lamp sizes shall be in accordance with IEC 60662.

Diffusers and lenses for HPS luminaries shall be glass.

4.5 Emergency Lighting

Emergency lighting shall be provided for the following types of areas:

- (1) Control rooms
- (2) Turbine rooms (escape lighting only)
- (3) Enclosed HRSG galleries and operating floors (escape lighting only)
- (4) Diesel and battery rooms
- (5) Electrical equipment relay and switchgear rooms
- (6) Cable spreading rooms (escape lighting only)
- (7) Chlorination room and hydrogen plant room (if applicable)
- (8) Locker, toilet and store rooms (escape lighting only)
- (9) All normally inhabited areas (escape lighting only)

Power Plant Area

Within the power plant area (HRSG room, turbine room, auxiliary bay and control room), a total of 7.5% of the total kilowatt lighting load shall be fed from an essential service bus.

Additional emergency lighting in these areas shall be provided by incandescent luminaries operated from the DC 220 volt station battery. The DC lighting shall be operated from automatic contactors which shall turn the lights on upon failure of all AC lighting in an area and turn the lights off upon the restoration of AC power. It shall also be possible to turn the AC lights on manually. The total dc lighting load shall be 2.5% of the total kilowatts of normal lighting.

The control room emergency lighting shall be as follows:

- (1) All of the AC lighting in the central control room shall be fed from the essential service buses.
- (2) DC lighting shall be provided in the control room to give an illumination level of at least 75 lux on the control boards and other areas which might require operator attention in an emergency.

Outlying Buildings

Emergency lighting in the areas not mentioned in the first paragraph for Power Plant Area shall be provided by self-contained luminaries. Each luminaire shall be a self-contained unit complete with storage battery, battery charger, transfer switch, battery voltage level indicator, low battery voltage alarming means and test switch.

Emergency lighting luminaries shall meet the requirements of IEC 60598 and as specified herein. Luminaries shall have an operating time of three (3) hours on self-contained battery power.

Each luminaire shall be designed to transfer from normal to emergency (battery) power upon loss of AC supply and return to normal power when the AC supply is restored. The emergency light shall be turned-on with emergency power and turned off with the restoration of the normal AC supply. A test switch in the emergency light pack shall simulate the power transfer function.

The battery charger shall maintain the batteries in a fully charged condition. Maintenance free sealed type batteries shall be provided. The batteries shall have a minimum life expectancy of four (4) years of normal operation.

The low battery voltage alarm shall be an intermittently operated buzzer and/or flashing indicating light.

The Contractor shall provide bilingual (exit/keluar) exit signs throughout buildings and stair-ways. The exit signs in enclosed areas shall be suitable for operation from the normal AC lighting source and for emergency operation from battery operated emergency lighting units.

All exits and paths to the exits shall be illuminated with emergency battery-operated luminaries. Equipment in outlying buildings that may be operated during an emergency shall also be illuminated in this manner.

Emergency and exit sign lighting luminaries located in areas classified as hazardous shall be suitable for the environment.

4.6 Aircraft Obstruction Lighting

4.6.1 General

The Contractor shall provide a complete obstruction lighting system in accordance with Indonesian Transportation Department (Departemen Perhubungan) requirements for bypass stack and HRSG stack. The system shall be accessible for maintenance from walkways and exterior platforms. Obstruction lighting shall be feed from essential bus to avoid loss of supply when fault happened.

4.6.2 Light Units

The detail of light units, illumination and location of light shall be in accordance with regulation by “Departemen Perhubungan”.

Failure of one (1) light shall not affect the others in the system.

All light units shall flash simultaneously at a rate of 20-60 flashes per minute.

4.6.3 Enclosures

All enclosures shall have a degree of protection of IPW 55 and shall be compatible in all respects with the connecting conduit. Enclosures within 16 meters of the top of each stack shall be of type 304 stainless steel.

4.6.4 Maintenance

It shall be possible to maintain an individual light unit or power supply unit while maintaining the remainder of the obstruction lighting system in service.

4.6.5 Wiring

All wiring shall be PVC insulated copper, and shall comply with the latest IEC requirements. All wire quantities and sizes shall be determined by the Contractor, based on a ambient temperature of 50°C.

4.6.6 Temporary Obstruction Lighting

The Contractor shall provide temporary Aeronautical Obstruction Lighting on the stack, during the progress of construction.

The Contractor shall furnish and install a fused connection to the temporary source.

It shall be the responsibility of the Contractor to notify immediately the nearest Departemen Perhubungan office facsimile of any extinguishment or improper flashing of the top level or temporary lights or light units that will last more than 30 minutes. Notification should be given immediately upon resumption of normal operation of the system.

4.7 Lighting Power Distribution Panels

The Contractor shall provide lighting power distribution panels, as required, for the lighting specified as part of the scope of work for this Contract and with 25% excess capacity and spare breakers for future use.

Each lighting power distribution panel shall be provided with a minimum of 18 single-pole and four (4) double-pole (for 400 V lighting) branch circuit breakers. Lighting power distribution panels shall meet the requirements of E-001.

Branch circuit breakers shall be automatic thermal magnetic type molded case breakers with an interrupting rating of 25 kA at 400 volts for AC lighting panels and 20 kA at 250 volts for DC lighting panels. The short circuit current at AC lighting panels may be limited to 25 kA by the use of current limiting method which shall be subject to the review of the Owner. All branch circuit breakers shall be equipped with both thermal and magnetic trip devices. (Refer to Technical Specification E-001).

Single-pole breakers shall be limited to approximately 2,200 watts load at 220 V and double-pole breakers shall be limited to approximately 5,000 watts load at 400 volts.

All buses and connections shall be hard-drawn or cold-rolled copper bars. The neutral bus shall be insulated from the panel enclosure. The load between the three (3) phases shall be balanced.

All insulating material used for buses, barriers or for enclosing conducting parts shall be moisture-resistant materials.

Panel enclosures shall be rigidly constructed of galvanized sheet steel in

accordance with NEMA PB 1. The thickness of the sheet steel shall be two (2) mm or more.

Each lighting panel shall have hinged doors which shall be provided with a lock and spring catch with suitable handles for releasing the catch. The panels shall be provided with wire ways at least 100 mm wide on all four (4) sides of each panel.

Each panel shall be provided with branch circuit numbers and service directory, to be attached to the inside of the door.

Each panel shall be provided with a bilingual (Indonesian and English) white nameplate with black engraving. The nameplate shall indicate the lighting cabinet number as per Owner's criteria. The lettering size shall be a minimum of ten (10) mm high.

All lighting panels shall be painted with two (2) coats both primer and finish coat of epoxy base paint. The Owner will specify the finish color later.

Enclosures for the lighting panels located in the electrical equipment rooms indoors shall have a degree of protection of IP 42. Lighting panel enclosures located any place other than electrical equipment rooms shall have a degree of protection of IPW 55.

4.8 Wire Size

The cable and wire for lighting circuits shall be sized so that the voltage drop from the output terminal of the transformer power source shall be limited to 1 % in main feeds to the lighting distribution panels and 2% in branch circuits.

The minimum wire size for lighting circuits shall be 2.5 mm².

4.9 Receptacles

The Contractor shall provide 230 V receptacles throughout the areas in his scope of work as specified herein.

All receptacles shall be fed from the lighting panels.

(1) Indoor Non-hazardous Areas

Receptacles shall be provided in all offices, kitchen areas, locker rooms, toilets, control rooms and electrical equipment rooms. The receptacles

shall be spaced at three (3) meter intervals measured along the walls. These receptacles shall conform to IEC 60083 design. Receptacles in locker rooms and toilets shall be provided with ground fault interrupters (residual current circuit breakers) with a sensitivity of not more than 30 mA (10mA or less preferred) meeting the requirements of IEC 60755.

(2) Indoor Hazardous and Outdoor Areas

Receptacles shall be provided in all other areas not included in Clause 4.9.(1). The receptacles shall be in accordance with IEC 60309 and shall be water-tight and dust-tight with hinged or screwed covers meeting the requirements of IPW 55. The receptacles shall be rated 250 volt, 16 ampere. All receptacles shall be of the interlocking type with mechanical switching or other suitable means, whereby, only after insertion of the plug the switch can be activated. It shall not be possible to pull the plug out, without first de-energizing the receptacle.

All receptacles located in hazardous areas shall be in accordance with IEC 60079 and NFPA 70 (NEC) Article 500.

One receptacle per 20 square meter area shall be provided. In addition, a 230 volt receptacle shall be conveniently located near each HRSG access door.

All receptacles located in hazardous areas shall be in accordance with IEC 60079 and NFPA 70 (NEC) Article 500, and receptacles in hazardous area.

4.10 Welding Receptacles

Welding receptacles shall be furnished and installed so that they are within convenient reach of a 30-meter extension cord in all areas except those areas listed in Clause 4.9.(1). Additional welding receptacles shall also be located within convenient reach of all electrically operated hoists and trollies or other portable equipment, as applicable.

In addition, a welding receptacle shall be conveniently located near each HRSG platform entrance, substation and underground tunnel entrances and manholes.

All welding receptacles shall comply with the requirements of IEC 60309. They shall be water-tight and dust-tight meeting the requirements for a degree of protection of IPW 55.

All welding receptacles shall be of the interlocking type with mechanical

switching whereby only after complete insertion of the appropriate plug, the switch can be activated. It shall not be possible to pull the plug out, unless the receptacle is de-energized. All welding receptacles shall be fused.

The welding receptacles shall be three (3) poles plus neutral, plus ground and shall be rated 63 amp at 400 V operating voltage, 3 phase, 50 Hz.

The Contractor shall furnish a total of 10 plugs suitable for the receptacles and fifteen (15) connectors with mechanical switching means.

Plugs and connectors shall be suitable for cable sizes ranging from 10 mm² (minimum) to 35 mm² (maximum). The receptacles shall be suitable for cable sizes ranging from 25 mm² (minimum) to 70 mm² (maximum). The cables shall be considered as having four (4) conductors with PVC insulation and suitable jacketing material.

4.11 Installation

Lighting luminaries and accessories shall be installed so that:

- (1) The luminaries will provide the full illumination level for the area in accordance with Table 4.1-1 Clause 4.1.
- (2) They are accessible for maintenance and changing of lamps.
- (3) They will not interfere with the installation of other equipment.
- (4) Other equipment or building components will not cause shadows or unacceptable levels of illumination in the illuminated area.

Luminaries mounted on metal siding shall be provided with a five (5) mm thick aluminum backing plate.

Lighting installations in hazardous areas shall comply with the requirements of NFPA 70 for the hazardous substance or substances in the area.

All luminaire outlet boxes not mounted directly from steel beams or concrete slabs shall be mounted with steel angles, steel channels, or manufactured steel framing supports or brackets welded or bolted to adjacent structural steel or walls. The Contractor shall furnish and install all such necessary supports.

Mounting Heights

The mounting heights of outlet boxes for switches, etc. from the finished floor to the centerline of the box shall be as follows:

Switches	1400 mm	Power outlets	460 mm
Thermostats	1400 mm	Clock outlets	2,300 mm
Receptacles	460 mm		

The dimensions from the finish floor to the top of lighting distribution panels shall be as follows:

<u>Panel Height</u>	<u>Top of Panel Above Floor</u>
Up to 1,200mm	1,800mm
Over 1,200 mm	2,000 mm

Conduit and Fittings

Conduit and fittings and their installation for lighting systems shall be in accordance with E-232.

All wiring for lighting equipment in hazardous areas, such as the battery rooms and the hydrogen plant building (if applicable), shall be installed using rigid conduit.

All lighting equipment in the following areas shall be wired using rigid steel conduit:

- (1) Points where conduits emerge from walls or floors.
- (2) Conduit sleeves through walls and floors.
- (3) Exposed outdoor areas such as the circulating water pump house, and unenclosed HRSG galleries.
- (4) Under the roof of high bay areas.

All conduit boxes under the roof of a high bay area shall be galvanized cast boxes with threaded hub connections.

Wiring Installation

All lighting system wires installed in conduits shall be installed with 600 volt, 90°C PVC insulation. A common neutral wire shall be used with each group of three (3) lighting circuits on opposite phase buses on three-phase, 4 wire,

400/230 volt systems.

All receptacle circuits or mixed circuits containing receptacle outlets and motor feeds shall be wired with a common neutral feed for each circuit.

For incandescent fixture connections, the light blue or grounded neutral wire shall be connected to the shell terminal of the lamp socket.

Switch leg drops of 10 mm² wire circuits shall be reduced to 4 mm² wire to facilitate connections to switch device terminal screws.

Conductors shall be color coded as follows:

- | | | | |
|-----|----------------------------|---|-----------------------|
| (1) | Neutral | : | Light Blue |
| (2) | Ungrounded Phase Conductor | : | Red, Yellow and Black |
| (3) | Ground Conductor | : | Green/Yellow |

The wire to be used for wiring incandescent luminaries from the outlet box to the lamp socket shall be 600 volt silicone rubber insulated flexible stranded wire in the following sizes:

- (1) Use 2.5 mm² for all luminaries up to and including 300 Watts.
- (2) Use 4 mm² for all luminaries of 500 Watts and above.
- (3) Exceptions may be made to the above when the luminaire is prewired.

The wire to be used for wiring fluorescent industrial and strip luminaries from the outlet box to wiring channel interior connections shall be three (3) cores heavy duty rubber cord. The Contractor shall use 2.5 mm² wire for individual fixtures and 4 mm² wire for drops to continuous run luminaries.

5. PERFORMANCE TESTING AND GUARANTEES

Routine and type testing for the lighting equipment shall be in accordance with the applicable standards listed in Clause 2 herein.

After completion, the Contractor shall test all luminaries and all receptacles to ensure normal operation.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control for the lighting equipment shall conform to the requirements of Part 4, Section-2, Subsection 2.7. The procedures as specified shall include, but not be limited to, the following:

- (1) Continuity and wiring tests
- (2) Packaging and shipping
- (3) Welding
- (4) Cleaning and painting

The documentation shall include, but not be limited to, the following:

- (1) Shop Test Reports
- (2) Inspection Reports
- (3) The Contractor prove that the airborne obstruction light system is in accordance with Departemen Perhubungan requirements in latest issue.

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